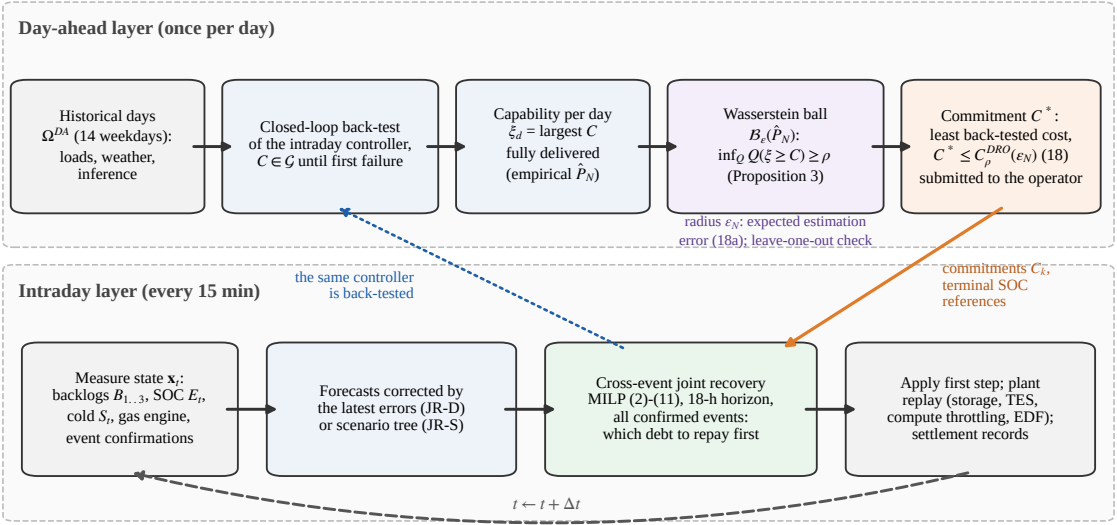
Day-ahead layer (once per day)
Historical days Ω^{DA} (14 weekdays): loads, weather, inference
Closed-loop back-test of the intraday controller, $C \in \mathcal{C}$ until first failure
Capability per day ξ_d = largest C fully delivered (empirical $\hat{P}_N$)
Wasserstein ball $\mathcal{B}_\varepsilon(\hat{P}_N)$: $\inf_Q Q(\xi \geq C) \geq \rho$ (Proposition 3)
Commitment C^*: least back-tested cost, $C^* \leq C_\rho^{DRO}(\varepsilon_N)$ (18) submitted to the operator
radius ε_N: expected estimation error (18a); leave-one-out check
the same controller is back-tested
commitments C_k, terminal SOC references
Intraday layer (every 15 min)
Measure state $\mathbf{x}_t$: backlogs $B_{1..3}$, SOC E_t, cold S_t, gas engine, event confirmations
Forecasts corrected by the latest errors (JR-D) or scenario tree (JR-S)
Cross-event joint recovery MILP (2)-(11), 18-h horizon, all confirmed events: which debt to repay first
Apply first step; plant replay (storage, TES, compute throttling, EDF); settlement records
$t \leftarrow t + \Delta t$